

QUESTION BANK

Description	
Course Code – Course Name	23OMG971-Total Quality Management
Semester	IV/ VII
Department	COMMON COURSE FT,CIVIL,AUTO,CSE,IT
Regulation	2023

Unit 1:

QNo	Questions	COs	Bloom's Level
Part A			
1.	State the meaning of Total Quality Management in the context of organizational excellence.	CO1	K1
2.	Mention the views of Deming and Juran on quality.	CO1	K1
3.	List the dimensions of product and service quality.	CO1	K1
4.	Differentiate between quality of design and quality of conformance.	CO1	K2
5.	Discuss how cost of quality can influence organizational decisions.	CO1	K2
6.	Interpret the concept of performance measures in the context of TQM implementation.	CO1	K2
7.	Interpret the significance of TQM in improving organizational performance.	CO1	K2
8.	List the benefits of TQM with respect to employee morale and productivity.	CO1	K1
9.	Compare the role of quality during the industrial revolution and the post-WWII era.	CO1	K2
10.	Interpret between reliability and responsiveness in the context of service quality.	CO1	K2
11.	List two common obstacles in successful TQM deployment in organizations.	CO1	K1
12.	List the basic components that form the foundation of a TQM framework.	CO1	K1
13.	List the stages through which quality evolved historically.	CO1	K1
14.	State the key enablers of Quality 4.0 in the context of real-time quality monitoring.	CO1	K1
15.	Interpret the significance of involving suppliers as strategic partners in a TQM environment.	CO1	K2
16.	List any two key features of Quality 4.0.	CO1	K1
17.	Identify the four eras in the evolution of quality.	CO1	K1
18.	List the six basic concepts of TQM mentioned in the framework.	CO1	K1

19.	Identify the three elements of Juran's quality trilogy.	CO1	K1
20.	List any two organizational benefits achieved through TQM implementation.	CO1	K1
Part – B			
1.	Outline the need for quality in modern organizations and trace the evolution of quality practices from inspection to Total Quality Management with suitable examples.	CO1	K2
2.	Explain Total Quality Management and critically discuss the six basic concepts that drive effective quality execution in organizations.	CO1	K2
3.	Explain in detail the dimensions of product quality with suitable examples.	CO1	K2
4.	Describe the TQM framework with a neat diagram and explain each component.	CO1	K2
5.	Explain the contributions of gurus of TQM to quality management.	CO1	K2
6.	Discuss the barriers to successful implementation of TQM in an organization and suggest suitable remedies	CO1	K2
7.	Explain in detail the dimensions of service quality with suitable examples	CO1	K2
8.	Explain the benefits of implementing TQM for customers, employees, organizations, and society.	CO1	K2

Unit 2:

QNo	Questions	COs	Bloom's Level
Part A			
1.	List any two principles of Deming's Philosophy.	CO2	K1
2.	Differentiate between a vision statement and a quality policy.	CO2	K2
3.	List the tools used for handling customer complaints.	CO2	K1
4.	Illustrate how customer service adds value in TQM.	CO2	K2
5.	Interpret how customer retention benefits the organization.	CO2	K2
6.	Name the five levels in Maslow's hierarchy of needs.	CO2	K1
7.	Illustrate how motivation impacts employee involvement in quality.	CO2	K2
8.	List two outcomes of empowering employees in TQM.	CO2	K1
9.	Label any two types of teams used in quality improvement.	CO2	K1

10.	State two functions of a quality council.	CO2	K1
11.	List two components of Juran's quality trilogy.	CO2	K1
12.	Match the four steps of the PDCA cycle with their purposes.	CO2	K1
13.	Outline the five elements of 5S.	CO2	K2
14.	List two key features of Kaizen.	CO2	K1
15.	Name two customer feedback methods used in TQM.	CO2	K1
16.	Select two elements that contribute to effective partnering.	CO2	K1
17.	List any two types of supplier sourcing strategies.	CO2	K1
18.	Interpret how supplier rating helps in quality management.	CO2	K2
19.	List two examples of recognition and reward in quality initiatives.	CO2	K1
20.	State the importance of supplier partnership in TQM.	CO2	K1
Part - B			
1.	Discuss Deming's philosophy and examine its role in driving continuous quality improvement.	CO2	K2
2.	Describe the role of leadership in TQM and explain the functions of the Quality Council.	CO2	K2
3.	Explain the components of quality statements and discuss the steps involved in strategic planning.	CO2	K2
4.	Describe customer perception of quality and explain the importance of customer feedback and complaint management.	CO2	K2
5.	Explain the Kano Model and discuss various customer retention strategies adopted by organizations.	CO2	K2
6.	Describe the concepts of motivation, empowerment, teamwork, recognition & reward, and performance appraisal in TQM.	CO2	K2
7.	Explain the three components of the Juran Trilogy and discuss their role in improving organizational quality performance.	CO2	K2
8.	Explain the PDCA cycle, 5S methodology, and Kaizen philosophy and discuss their importance in continuous process improvement.	CO2	K2

Unit 3:

QNo	Questions	COs	Bloom's Level
Part A			

1.	Compare the seven traditional quality tools and identify the situations in which each tool is most suitable.	CO3	K2
2.	Illustrate how the DMAIC methodology contributes to continuous quality improvement in manufacturing industries.	CO3	K2
3.	Interpret the significance of Process Capability in achieving customer satisfaction.	CO3	K2
4.	Compare Benchmarking and Six Sigma in terms of their objectives and organizational benefits.	CO3	K2
5.	Outline the stages involved in the Benchmarking process and discuss their significance.	CO3	K2
6.	Illustrate the intent and documentation requirements of Failure Mode and Effects Analysis (FMEA).	CO3	K2
7.	Differentiate between Design FMEA and Process FMEA with suitable examples.	CO3	K2
8.	Interpret how a Pareto chart assists managers in prioritizing quality improvement activities.	CO3	K2
9.	Illustrate the role of Cause-and-Effect diagrams in identifying quality-related problems.	CO3	K2
10.	Infer the importance of learning from Benchmarking results before implementing improvements.	CO3	K2
11.	Compare the applications of Histograms and Control Charts in process monitoring.	CO3	K2
12.	Outline the objectives of Six Sigma in improving process performance.	CO3	K2
13.	State the importance of Process Capability indices in quality management.	CO3	K1
14.	Relate the use of Scatter Diagrams to identifying relationships between process variables.	CO3	K2
15.	List the benefits of implementing Benchmarking in manufacturing organizations.	CO3	K1
16.	Classify the traditional quality tools based on their applications in quality improvement.	CO3	K2
17.	Interpret the importance of Risk Priority Number (RPN) in FMEA.	CO3	K2
18.	Compare internal, competitive and functional Benchmarking with suitable examples.	CO3	K2
19.	Illustrate the significance of Control Charts in monitoring process variation.	CO3	K2
20.	Infer the contribution of new management tools to continuous quality improvement in organizations.	CO3	K2
Part – B			
1.	Discuss the role of the seven traditional quality tools in quality improvement and illustrate their applications with suitable examples.	CO3	K2

2.	Compare the traditional quality tools and new management tools used in Total Quality Management, highlighting their objectives and applications.	CO3	K2
3.	Interpret the DMAIC methodology of Six Sigma and discuss its contribution to process improvement and customer satisfaction.	CO3	K2
4.	Discuss the concept of Process Capability and illustrate the importance of process capability indices in quality management.	CO3	K2
5.	Describe the Benchmarking process and interpret the significance of each stage in achieving organizational excellence.	CO3	K2
6.	Compare internal, competitive, functional and generic Benchmarking, and discuss their advantages with suitable industrial examples.	CO3	K2
7.	Discuss the objectives, documentation requirements and stages involved in Failure Mode and Effects Analysis (FMEA).	CO3	K2
8.	Differentiate Design FMEA and Process FMEA, and illustrate their significance in improving product and process quality.	CO3	K2

Unit 4:

Q.No	Questions	CO's	Bloom's Level
Part A			
1.	State the concept of Quality Circle.	CO4	K1
2.	Outline the purpose of Quality Circles in an organization.	CO4	K2
3.	State the concept of Quality Function Deployment (QFD).	CO4	K1
4.	Interpret how Quality Function Deployment (QFD) translates customer requirements into technical requirements.	CO4	K2
5.	Illustrate the role of the House of Quality in product development.	CO4	K2
6.	State the concept of Taguchi Quality Loss Function.	CO4	K1
7.	Interpret the significance of Taguchi Quality Loss Function in quality improvement.	CO4	K2
8.	Write the mathematical expression of Taguchi Quality Loss Function.	CO4	K1
9.	State the concept of Total Productive Maintenance (TPM).	CO4	K1
10.	Outline the need for Total Productive Maintenance (TPM) in organizations.	CO4	K2

11.	Discuss the role of the pillars of TPM in improving equipment effectiveness.	CO4	K2
12.	Interpret the importance of continuous improvement in quality management.	CO4	K2
13.	State the concept of Performance Measure.	CO4	K1
14.	Compare financial and non-financial performance measures.	CO4	K2
15.	State the concept of Cost of Quality (COQ).	CO4	K1
16.	Classify the categories of Cost of Quality (COQ).	CO4	K2
17.	Interpret the importance of Cost of Quality (COQ) in reducing organizational costs.	CO4	K2
18.	State the concept of Business Process Reengineering (BPR).	CO4	K1
19.	Differentiate Business Process Reengineering (BPR) from Continuous Improvement.	CO4	K2
20.	Compare the traditional quality concept with Taguchi's quality concept.	CO4	K2
Part – B			
1.	An organization experiences low employee participation in solving quality-related problems. Apply the concept of Quality Circles to improve employee involvement and organizational performance.	CO4	K3
2.	A company receives frequent customer complaints because customer requirements are not adequately reflected in product design. Apply Quality Function Deployment (QFD) and the House of Quality to translate customer requirements into technical specifications.	CO4	K3
3.	A manufacturing organization experiences frequent machine breakdowns, excessive downtime, and reduced productivity. Apply the principles of Total Productive Maintenance (TPM) to improve equipment effectiveness and operational efficiency.	CO4	K3
4.	A manufacturing process consistently produces products within specification limits, yet customers report dissatisfaction with quality. Apply Taguchi's Quality Loss Function to interpret the quality loss and recommend suitable improvement measures.	CO4	K3
5.	A company evaluates its performance only through financial indicators. Apply appropriate financial and non-financial performance measures to develop a balanced performance measurement system.	CO4	K3
6.	An organization incurs high prevention, appraisal, internal failure, and external failure costs. Apply the Cost of Quality (COQ) framework to classify these costs and recommend strategies for reducing the overall cost of	CO4	K3

	quality.		
7.	An organization experiences process delays, high operational costs, increasing customer complaints, and inefficient workflows. Apply the principles of Business Process Reengineering (BPR) to redesign the process and improve organizational performance.	CO4	K3
8.	An organization aims to improve customer satisfaction, operational efficiency, product quality, and overall organizational performance. Apply the concepts of Quality Circles, QFD, TPM, Cost of Quality (COQ), and Business Process Reengineering (BPR) to propose an integrated quality improvement strategy.	CO4	K3

Unit 5:

Q.No	Questions	CO's	Bloom's Level
Part A			
1.	State the objectives of the ISO 9000 series of standards.	CO5	K1
2.	List the mandatory documented information required in ISO 9001.	CO5	K1
3.	Interpret the importance of ISO 9001 certification for manufacturing organizations.	CO5	K2
4.	Illustrate the purpose and objectives of internal quality audits.	CO5	K2
5.	Differentiate between first-party, second-party and third-party audits.	CO5	K2
6.	Interpret the steps involved in implementing ISO 9001 in an organization.	CO5	K2
7.	Illustrate the procedure for conducting an internal quality audit using a flowchart.	CO5	K2
8.	Outline the implementation plan for obtaining ISO 9001 certification in an organization.	CO5	K2
9.	Recommend suitable sector-specific ISO standards (AS9100, IATF 16949 and TL9000) for different industries with justification.	CO5	K3
10.	Prepare an audit checklist for evaluating the Quality Management System of an organization.	CO5	K3
11.	Define Environmental Management System (EMS).	CO6	K1
12.	State the requirements of ISO 14001.	CO6	K1
13.	Interpret the planning requirements of ISO 14001.	CO6	K2
14.	Illustrate the checking and corrective action elements of ISO 14001.	CO6	K2
	Compare ISO 9001 and ISO 14001 with respect to objectives, scope and implementation.	CO6	K2

15.	Interpret the role of an Environmental Management System in supporting sustainable industrial development.	CO6	K2
16.	Recommend suitable Environmental Management System practices based on ISO 14001 to minimize environmental impacts in an organization.	CO6	K3
17.	Illustrate the Environmental Management System framework for an organization seeking ISO 14001 certification.	CO6	K2
18.	Analyze the environmental and organizational benefits of implementing ISO 14001 in an organization.	CO6	K4
19.	Analyze how internal environmental audits contribute to continual improvement and regulatory compliance in an organization.	CO6	K4
Part – B			
1.	Develop a Quality Management System implementation strategy in a manufacturing organization. Illustrate the documentation, implementation phases, resource allocation, and continual improvement process.	CO5	K3
2.	Examine a quality audit plan for an organization by selecting appropriate types of quality audits. Include the audit schedule, audit methodology, audit criteria, reporting procedures, and follow-up actions.	CO5	K4
3.	Analyse the implementation plan for obtaining ISO 9001 certification in an organization. Include the key requirements, documentation, implementation stages, and expected organizational benefits.	CO5	K3
4.	Apply suitable sector-specific quality standards (AS9100, IATF 16949, and TL9000) to different industries and justify their selection based on organizational requirements.	CO5	K3
5.	Develop an Environmental Management System (EMS) implementation strategy based on ISO 14001 for an organization. Include environmental policy, planning, implementation, performance evaluation, and continual improvement.	CO6	K3
6.	Design an implementation and registration plan for obtaining ISO 14001 certification in an organization. Include documentation, internal audits, certification procedures, and continual improvement activities.	CO6	K4
7.	Apply the requirements of ISO 14001 to develop an Environmental Management System that addresses environmental aspects, legal compliance, operational control, and monitoring.	CO6	K3
8.	Analyze the organizational and environmental benefits of implementing ISO 14001 and recommend suitable Environmental Management System practices for achieving sustainable organizational performance.	CO6	K4